CondAstro-Miner: Disentangling Astrophysical Anomalies from Observational Artifacts in Massive Sky Surveys via Invariant Conditional Pattern Mining

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Abstract

The explosion of wide-field astronomical sky surveys (Gaia, SDSS, ZTF, and the upcoming Vera C. Rubin Observatory) has catalyzed intense interest in unsupervised machine learning for discovering rare astrophysical phenomena. However, conventional anomaly detection algorithms (e.g., Isolation Forest, Local Outlier Factor, One-Class SVM, and deep autoencoders) evaluate objects in an unconditional, global feature space. Consequently, empirical audits reveal that over 85% of top-ranked anomalies are not novel astrophysical discoveries, but spurious observational artifacts, including faint-end Malmquist bias, severe interstellar extinction near the Galactic plane, crowded-field blending, and reduction defects. To overcome this fundamental bottleneck, we propose **CondAstro-Miner**, a novel pattern mining framework grounded in invariant causal principles. Rather than asking whether an object is globally isolated, CondAstro-Miner discovers *conditional contrast patterns* across partitioned observational strata (signal-to-noise regimes, Galactic latitudes, and astrometric uncertainty bands). We formulate the *Invariance Contrast Ratio* (ICR), a mathematically sound scoring function that separates context-dependent artifacts (which appear anomalous only under localized, noisy observational regimes) from true physical anomalies (which preserve anomaly invariance across independent observing environments). Furthermore, we establish a pruning bound that guarantees complete, exact search without exhaustive evaluation. Evaluated on 500,000 stellar objects from Gaia Data Release 3 with injection benchmarks and spectroscopic cross-matching, CondAstro-Miner reduces the False Discovery Rate from 88.4% (Isolation Forest) down to 8.2%, achieves an Invariance Rank Correlation of $\rho = 0.89$, and automatically extracts compact, human-interpretable symbolic rules that explain the physical or instrumental origin of every flagged subgroup.

Keywords: Astronomical Anomaly Detection, Conditional Pattern Mining, Invariant Prediction, Observational Artifacts, Gaia DR3, Subgroup Discovery.

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1 Introduction

Wide-field photometric, astrometric, and spectroscopic sky surveys represent the primary engine of modern astronomical discovery. Missions such as the European Space Agency's *Gaia* astrometric satellite [Gaia Collaboration et al., 2023], the Sloan Digital Sky Survey (SDSS), the Dark Energy Survey (DES), the Zwicky Transient Facility (ZTF), and the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) catalogue billions of celestial objects. Within these vast data volumes lie extraordinarily rare astrophysical specimens—hypervelocity stars fleeing the Galactic center, peculiar white dwarf-main sequence binaries, exotic cataclysmic variables, and transitional supernovae [Baron and Poznanski, 2017, Pruzhinskaya et al., 2019, Malanchev et al., 2021]. Because the exact observational signatures of unknown phenomena cannot be prespecified in a training set, the discovery of novel physics fundamentally relies on *unsupervised anomaly detection* [Lochner and Bassett, 2021, Baron, 2019].

1.1 The Crisis of Spurious Outliers in Astronomical Machine Learning

Despite extensive algorithmic experimentation ranging from classical Isolation Forests [Giles and Walkowicz, 2019] and Random Forest distance metrics [Baron and Poznanski, 2017, Reis et al., 2018] to Deep Variational

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Autoencoders [Villar et al., 2020] and Generative Adversarial Networks [Prochaska et al., 2019], standard machine learning pipelines suffer from an acute methodological failure: **the dominance of observational and selection artifacts**.

In unconditional outlier mining, an algorithm assigns an anomaly score based purely on an object's isolation or low local density in a feature space $\mathcal{X}$ (e.g., colors, absolute magnitudes, proper motions, parallax). However, astronomical measurements are inherently heterogeneous, heteroscedastic, and non-uniformly corrupted [Baron, 2019, Kennamer et al., 2020]:

1. **Faint-End Measurement Degradation (Malmquist Bias):** Near the survey detection limit (e.g., Gaia magnitude $G > 19.5$), photon shot noise escalates exponentially, driving parallax uncertainties $\sigma_\varpi/\varpi > 0.5$. In an unconditional feature space, these noisy objects naturally scatter into sparse regions, masquerading as hyper-luminous or hypervelocity candidates.
2. **Galactic Extinction and Stellar Crowding:** Towards low Galactic latitudes ($|b| < 10°$), dust reddening ($E(B-V)$) and overlapping point spread functions (PSF blending) severely distort apparent stellar colors and astrometry, mimicking intrinsically cool, exotic red stars.
3. **Instrumental and Pipeline Flaws:** Satellite thruster firings, cosmic ray hits, diffraction spikes, and sky-subtraction residuals create sharp, non-astrophysical feature vectors that dominate top-ranked outlier lists [Giles and Walkowicz, 2019, Narayan et al., 2018, Aleo et al., 2023].

Empirical audits across SDSS, ZTF, and Gaia confirm that over 85% of the highest-ranked outliers surfaced by unconditional algorithms represent unphysical measurement flaws rather than genuine astrophysical discoveries [Baron, 2019, Lochner and Bassett, 2021]. While active learning frameworks like Astronomaly [Lochner and Bassett, 2021, Ishida et al., 2021] mitigate this through iterative human-in-the-loop relabeling, human vetting cannot scale to the millions of candidates anticipated from LSST.

1.2 The Invariance Principle: Disentangling Physics from Artifacts

To resolve this crisis, we introduce an orthogonal paradigm: **Conditional Pattern Mining under Invariant Causal Principles** [Peters et al., 2016, Fournier-Viger et al., 2019]. We argue that an anomaly cannot be judged in isolation from its observational context. Instead, we separate the feature space into intrinsic physical attributes $\mathbf{x}$ and external observational conditions $\mathbf{c}$ (e.g., signal-to-noise regime, Galactic latitude, astrometric excess noise).

Under this formulation, we observe an essential epistemic dichotomy:

- **Observational Artifacts are Context-Dependent:** A spurious subgroup appears anomalous *only* under a specific, corrupted observational regime (e.g., only when $G > 19.5$ and astrometric noise is extreme), but vanishes when audited in high-SNR regimes.
- **Astrophysical Anomalies are Invariant:** A genuine physical population (e.g., bona fide hypervelocity stars or extreme subdwarfs) retains its anomalous departure across distinct, independent observational environments.

1.3 Contributions

In this paper, we propose **CondAstro-Miner**, an exact and interpretable conditional pattern mining algorithm tailored for astronomical survey discovery. Our specific contributions are:

1. **Formal Problem Definition:** We formalize the distinction between invariant physical anomalies and context-dependent observational artifacts through the *Invariance Contrast Ratio* (ICR).
2. **Tight Pruning Bound:** We prove Lemma 1, which provides a sound upper bound on the achievable ICR for recursive pattern extensions, enabling massive combinatorial search space pruning.
3. **Interpretable Rule Induction:** Unlike deep learning models, CondAstro-Miner outputs concise symbolic rules that explicitly identify whether a subgroup is driven by genuine physics or specific instrumental noise channels.
4. **Rigorous Empirical Audit:** We evaluate the algorithm on 500,000 Gaia DR3 sources against state-of-the-art baselines (Astronomaly, Isolation Forest, LOF, VAE), demonstrating an 80% reduction in False Discovery Rate and verified discovery of real rare stellar populations.

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2 Related Work

2.1 Unsupervised Anomaly Detection in Astronomy

Unsupervised outlier identification in astronomy has evolved from distance-based methods to manifold learning and deep generative models. Baron and Poznanski [2017] and Reis et al. [2018] pioneered the use of unsupervised Random Forest similarity matrices to discover peculiar quasars and supernovae in SDSS. Giles and Walkowicz [2019] applied Isolation Forests and DBSCAN to 170,000 Kepler light curves, revealing significant contamination from spacecraft momentum dumps. In the time domain, Villar et al. [2020] used Variational Autoencoders to encode ZTF light curves, while Prochaska et al. [2019] and Storey-Fisher et al. [2024] deployed GANs and normalizing flows to identify anomalous spectra and galaxy morphologies. However, as critically synthesized by Baron [2019], all unconditional methods conflate extreme noise in low-density feature space with true astrophysical novelty.

2.2 Active Learning and Human-in-the-Loop Filtering

To prevent astronomers from wasting telescope follow-up time on artifacts, Lochner and Bassett [2021] developed Astronomaly, an active learning tool that dynamically updates outlier rankings based on user scores. Ishida et al. [2021] and Malanchev et al. [2021] integrated active learning into ZTF transient brokers, while Čubrilo et al. [2024] explored self-supervised contrastive learning to build survey-invariant representations. While active learning incorporates domain expertise, it remains labor-intensive and susceptible to subjective confirmation bias.

2.3 Contrast Pattern Mining and Subgroup Discovery

In symbolic data mining, Contrast Pattern Mining [Fournier-Viger et al., 2019] and Subgroup Discovery [Duivesteijn et al., 2016] discover patterns whose frequency or statistical behavior differs significantly across predefined classes or environments. Exceptional Model Mining (EMM) [Duivesteijn et al., 2016] extends this to identify subsets of data where model relationships deviate from the global norm. In parallel, Peters et al. [2016] formalized Invariant Causal Prediction, establishing that causal mechanisms remain invariant across diverse experimental perturbations. CondAstro-Miner bridges these domains by applying invariant subgroup discovery directly to astronomical survey catalogs.

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3 Problem Formulation & Mathematical Foundation

3.1 Astronomical Catalog Model

Let $\mathcal{D} = \{O_1, O_2, \ldots, O_N\}$ be an astronomical catalog containing N celestial objects. Each object is defined as a tuple:

$$O_i = (\mathbf{x}_i, \mathbf{c}_i) \tag{1}$$

where:

- $\mathbf{x}_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,d}) \in \mathcal{X}$ is a d-dimensional vector of **intrinsic astrophysical attributes** (e.g., color indices $G_{\mathrm{BP}} - G_{\mathrm{RP}}$, absolute magnitude M_G, tangential velocity v_t, spectral line equivalent widths).
- $\mathbf{c}_i = (c_{i,1}, c_{i,2}, \ldots, c_{i,k}) \in \mathcal{C}$ is a k-dimensional vector of **observational and instrumental context parameters** (e.g., apparent magnitude G, Galactic latitude b, astrometric excess noise significance ϵ_i, signal-to-noise ratio SNR).

3.2 Observational Partitioning (Strata)

To assess invariance, the observational context space $\mathcal{C}$ is partitioned into a finite set of K mutually exclusive, exhaustive observational environments or strata:

$$\mathcal{E} = \{E_1, E_2, \ldots, E_K\}, \quad \bigcup_{k=1}^{K} E_k = \mathcal{D}, \quad E_j \cap E_k = \emptyset \text{ for } j \neq k \tag{2}$$

Representative strata include Galactic field density splits ($E_{\mathrm{plane}} : |b| < 10^\circ$ vs. $E_{\mathrm{halo}} : |b| > 50^\circ$) and signal-to-noise tiers ($E_{\mathrm{bright}} : G \leq 15$ vs. $E_{\mathrm{faint}} : G > 18.5$).

3.3 Local Anomaly Scoring

Let $\alpha \subseteq \mathcal{X}$ represent an itemset or attribute-value subgroup (e.g., $[v_t > 400 \text{ km/s}] \wedge [G_{\text{BP}} - G_{\text{RP}} < 0.2]$). For any observational stratum E_k, we define the local anomaly score $A(\alpha \mid E_k)$ as the standardized statistical divergence of subgroup α from the stratum background:

$$A(\alpha \mid E_k) \triangleq \frac{D_{\text{KL}}\big(P(\mathbf{x} \in \alpha \mid E_k) \,\|\, P(\mathbf{x} \mid E_k)\big)}{\sigma(E_k)} \tag{3}$$

where D_{KL} is the Kullback-Leibler divergence and $\sigma(E_k)$ normalizes for the intrinsic noise level of stratum E_k.

3.4 The Invariance Contrast Ratio (ICR)

Definition 1 (Mean Anomaly and Environmental Variance)**.** *For a pattern α, its mean anomaly score across all K strata is:*

$$\bar{A}(\alpha) \triangleq \frac{1}{K} \sum_{k=1}^{K} A(\alpha \mid E_k) \tag{4}$$

and its cross-environment variance is:

$$\text{Var}_{\mathcal{E}}(\alpha) \triangleq \frac{1}{K} \sum_{k=1}^{K} \Big(A(\alpha \mid E_k) - \bar{A}(\alpha) \Big)^2 \tag{5}$$

Definition 2 (Invariance Contrast Ratio)**.** *The Invariance Contrast Ratio (*ICR*) of pattern α across environment set $\mathcal{E}$ is formulated as:*

$$\text{ICR}(\alpha) \triangleq \frac{\bar{A}(\alpha)}{\sqrt{\text{Var}_{\mathcal{E}}(\alpha)} + \epsilon} \tag{6}$$

where $\epsilon > 0$ is a small smoothing constant preventing division by zero.

Definition 3 (Invariant Physical Anomaly vs. Observational Artifact)**.** *Given a minimum anomaly threshold τ_{anom} and an invariance threshold θ_{inv}:*

- *Pattern α is an **Invariant Physical Anomaly** if:*

$$\bar{A}(\alpha) \geq \tau_{anom} \quad \textit{and} \quad \text{ICR}(\alpha) \geq \theta_{inv} \tag{7}$$

- *Pattern α is an **Observational Artifact Subgroup** if:*

$$\bar{A}(\alpha) \geq \tau_{anom} \quad \textit{and} \quad \text{ICR}(\alpha) < \theta_{inv} \tag{8}$$

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4 The Proposed Algorithm: CondAstro-Miner

4.1 Search Strategy and Upper Bound Pruning

Direct enumeration of all candidate subgroups $\alpha \subseteq \mathcal{X}$ is combinatorially explosive ($O(2^{|\mathcal{X}|})$). Because $\text{ICR}(\alpha)$ depends on both the mean and the variance across strata, it does not satisfy the downward-closure property (monotonicity).

To achieve exact and complete mining without heuristic approximations, we establish a tight upper bound on ICR.

Lemma 1 (Upper Bound on Invariance Contrast Ratio)**.** *Let α be a pattern and $\beta \supset \alpha$ be any specialized extension of α. Let $A_{\max}(\alpha \mid E_k) = \max_{O \in E_k, O \supseteq \alpha} score(O)$. Then:*

$$\bar{A}(\beta) \leq \bar{A}_{\max}(\alpha) \triangleq \frac{1}{K} \sum_{k=1}^{K} A_{\max}(\alpha \mid E_k) \tag{9}$$

Algorithm 1 CondAstro-Miner Main Mining Procedure

Require: Catalog $\mathcal{D}$, Feature space $\mathcal{X}$, Observational Strata $\mathcal{E} = \{E_1, \ldots, E_K\}$, Thresholds $\tau_{\text{anom}}, \theta_{\text{inv}}, \text{min_supp}$
Ensure: Set of Invariant Physical Anomalies $\mathcal{P}_{\text{phys}}$, Set of Artifact Rules $\mathcal{P}_{\text{art}}$

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Discretize continuous attributes 𝒳 into equispaced/quantile bins ℐ
𝒫_phys ← ∅; 𝒫_art ← ∅
ExploreSubtree(∅, ℐ, ℰ, τ_anom, θ_inv)
return 𝒫_phys, 𝒫_art
procedure ExploreSubtree(α, Candidates, ℰ, τ_anom, θ_inv)
    for all item ∈ Candidates do
        β ← α ∪ {item}
        if (β | 𝒟) < min_supp then continue
        end if
        Compute A(β | E_k) for all k ∈ {1, ..., K}
        Compute Ā(β), Var_ℰ(β), and ICR(β)
        if Ā(β) ≥ τ_anom then
            if ICR(β) ≥ θ_inv then
                𝒫_phys ← 𝒫_phys ∪ {β}                ▷ True Physical Anomaly
            else
                𝒫_art ← 𝒫_art ∪ {β}                  ▷ Observational Artifact Subgroup
            end if
        end if
        Compute UB_ICR(β) using Lemma 1
        if UB_ICR(β) ≥ θ_inv then
            NewCand ← {j ∈ Candidates | j ≻ item}
            ExploreSubtree(β, NewCand, ℰ, τ_anom, θ_inv)
        end if
    end for
end procedure
```

Furthermore, the maximum achievable ICR *for any extension of α is bounded by:*

$$\text{UB}_{ICR}(\alpha) \triangleq \frac{\bar{A}_{\max}(\alpha)}{\epsilon} \tag{10}$$

and if the minimum observed variance across children is bounded below by $V_{\min}(\alpha)$, then:

$$\text{UB}_{ICR}(\alpha) \triangleq \frac{\bar{A}_{\max}(\alpha)}{\sqrt{V_{\min}(\alpha)} + \epsilon} \tag{11}$$

Proof. By definition, for any extension $\beta \supset \alpha$, the supporting object set satisfies $T(\beta) \subseteq T(\alpha)$. Thus, the maximum local score of any element in $T(\beta)$ cannot exceed the maximum local score of $T(\alpha)$, yielding $A(\beta \mid E_k) \leq A_{\max}(\alpha \mid E_k)$. Summing over all K strata and dividing by K proves $\bar{A}(\beta) \leq \bar{A}_{\max}(\alpha)$. Since variance is non-negative ($\text{Var}_{\mathcal{E}}(\beta) \geq V_{\min}(\alpha) \geq 0$), the denominator satisfies $\sqrt{\text{Var}_{\mathcal{E}}(\beta)} + \epsilon \geq \sqrt{V_{\min}(\alpha)} + \epsilon$. Dividing the maximal numerator by the minimal denominator completes the proof. □

Theorem 1 (Completeness and Correctness)**.** *Pruning any search branch where $\text{UB}_{ICR}(\alpha) < \theta_{inv}$ introduces zero false dismissals, guaranteeing the discovery of all valid invariant physical anomalies.*

4.2 Algorithmic Structure

Algorithm 1 details the recursive depth-first exploration with invariant contrast pruning.

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5 Experimental Evaluation Protocol

5.1 Experimental Setup & Catalog Datasets

Experiments are conducted on a benchmark of 500,000 stellar sources extracted from **Gaia Data Release 3 (DR3)** [Gaia Collaboration et al., 2023], cross-matched with SDSS DR16 spectroscopy and 2MASS

infrared photometry.

- **Astrophysical Features ($\mathcal{X}$):** $G_{\mathrm{BP}} - G_{\mathrm{RP}}$ color, absolute magnitude M_G (computed via photogeometric distance), tangential velocity $v_t = 4.74047 \cdot \mu/\varpi$, and total proper motion $\mu = \sqrt{\mu_{\alpha*}^2 + \mu_\delta^2}$.
- **Observational Strata ($\mathcal{E}$):**
  1. Galactic latitude tiers: $E_{\mathrm{plane}}(|b| < 10°)$, $E_{\mathrm{intermediate}}(10° \leq |b| \leq 40°)$, and $E_{\mathrm{halo}}(|b| > 40°)$.
  2. Photometric SNR tiers: $E_{\mathrm{bright}}(G \leq 14)$, $E_{\mathrm{medium}}(14 < G \leq 18)$, and $E_{\mathrm{faint}}(G > 18)$.
  3. Astrometric quality indicator: Renormalized Unit Weight Error (RUWE < 1.4 vs. RUWE ≥ 1.4).

5.2 Baselines

We compare CondAstro-Miner against five established algorithms:

1. **Isolation Forest (iForest):** Standard scikit-learn implementation with 200 isolation trees.
2. **Local Outlier Factor (LOF):** Density-based nearest-neighbor anomaly scoring ($k = 20$).
3. **One-Class SVM (OC-SVM):** Non-linear kernel boundary estimator ($\gamma = \text{scale}$).
4. **Deep VAE:** Variational Autoencoder with 3-layer MLP encoder/decoder [Villar et al., 2020].
5. **Astronomaly:** Active anomaly detection framework using Isolation Forest with simulated human feedback [Lochner and Bassett, 2021].

5.3 Benchmark 1: Controlled Injection-Recovery Test

To establish ground truth, we inject $1,000$ synthetic physical anomalies (true hypervelocity stars with $v_t \in [500, 800]$ km/s and rare hot subdwarfs) and $10,000$ synthetic observational artifacts (faint-end noise spikes, severe extinction reddening, and astrometric wobbles).

Table 1 summarizes the comparative results across Top-100 anomaly candidates.

Table 1: Performance comparison on the Gaia DR3 Injection-Recovery Benchmark (Top-100 candidates). ARR: Artifact Rejection Rate; FDR: False Discovery Rate; Recall: True physical anomaly recovery.

Algorithm	Precision@100	Recall@100	ARR (%)	FDR (%)	PR-AUC
One-Class SVM	0.080	0.008	11.2	92.0	0.124
LOF	0.110	0.011	14.5	89.0	0.158
Isolation Forest	0.116	0.012	15.3	88.4	0.172
Deep VAE	0.190	0.019	26.8	81.0	0.285
Astronomaly (20 iterations)	0.540	0.054	61.0	46.0	0.589
CondAstro-Miner (Proposed)	**0.918**	**0.092**	**94.1**	**8.2**	**0.934**

Key Finding: Standard unconditional algorithms (iForest, OC-SVM, LOF) suffer from catastrophic False Discovery Rates ($> 88\%$), prioritizing spurious faint-end noise. CondAstro-Miner achieves 94.1% Artifact Rejection and slashes the FDR down to 8.2%.

5.4 Benchmark 2: Cross-Environment Invariance Stability

We evaluate the rank stability of top-ranked anomalies when evaluated across independent Galactic latitude environments (E_{plane} vs. E_{halo}). Table 2 reports Spearman's rank correlation (ρ) and Kendall's τ. **Key Finding:** While baseline anomaly rankings collapse across environments ($\rho < 0.32$, driven by shifting background stellar density), CondAstro-Miner maintains $\rho = 0.894$, proving mathematical robustness against selection effects.

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Table 2: Cross-Environment Invariance Metrics between Plane ($|b| < 10°$) and Halo ($|b| > 40°$).

Algorithm	Spearman's ρ	Kendall's τ	Invariance Drift (%)
Isolation Forest	0.182	0.125	81.8
Local Outlier Factor	0.141	0.098	85.9
Deep VAE	0.312	0.218	68.8
CondAstro-Miner	**0.894**	**0.781**	**10.6**

Table 3: Representative Subgroup Rules extracted by CondAstro-Miner on Gaia DR3.

Extracted Symbolic Rule	Classification	ICR	Support
$\mathbf{r_1}$: $[v_t > 450$ km/s$] \wedge [M_G \in (3, 7)] \wedge$ [Invariant $\forall b$]	Invariant Physical Anomaly	14.82	47
$\mathbf{r_2}$: $[G_{\mathrm{BP}} - G_{\mathrm{RP}} < -0.2] \wedge [M_G > 10] \wedge$ [Invariant $\forall b$]	Invariant Physical Anomaly	12.15	128
$\mathbf{r_3}$: $[v_t > 600$ km/s$] \wedge [G > 19.8] \wedge$ [RUWE $> 2.5] \wedge$ $[\lvert b\rvert < 5°]$	Observational Artifact	0.32	4,120
$\mathbf{r_4}$: $[G_{\mathrm{BP}} - G_{\mathrm{RP}} > 3.8] \wedge [E(B-V) > 1.2] \wedge [\lvert b\rvert < 8°]$	Observational Artifact	0.41	8,915

6 Astrophysical Interpretation & Explainable Rule Audit

A paramount advantage of CondAstro-Miner over deep neural architectures is its direct generation of symbolic, verifiable rules. Table 3 presents representative rules discovered from the Gaia DR3 catalog.
Astrophysical Validation:

- Rule $\mathbf{r_1}$ isolates true candidate hypervelocity runaway stars whose kinematic anomaly persists regardless of Galactic dust column density. Cross-matching with SDSS spectra confirms radial velocities exceeding 400 km/s.
- Rule $\mathbf{r_2}$ successfully isolates rare hot subdwarf and white dwarf populations with high confidence.
- Conversely, Rules $\mathbf{r_3}$ and $\mathbf{r_4}$ automatically bundle and neutralize thousands of false positives: $\mathbf{r_3}$ captures astrometrically astigmatic faint sources, while $\mathbf{r_4}$ flags interstellar extinction reddening.

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7 Conclusion

In this work, we addressed the fundamental failure of unconditional machine learning in astronomical discovery: the misclassification of measurement noise and selection biases as novel physics. We presented **CondAstro-Miner**, an exact pattern mining algorithm that formalizes the Invariance Principle across partitioned observational strata. By evaluating the Invariance Contrast Ratio (ICR) and pruning non-promising branches via a sound mathematical upper bound, CondAstro-Miner eliminates over 94% of observational artifacts while recovering genuine rare populations with 91.8% precision. The generated symbolic rules provide astronomers with immediate, physics-grounded explanations, establishing an automated, interpretable discovery engine ready for deployment on next-generation petascale surveys such as the Vera C. Rubin Observatory.

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